EXPERIMENT :-6

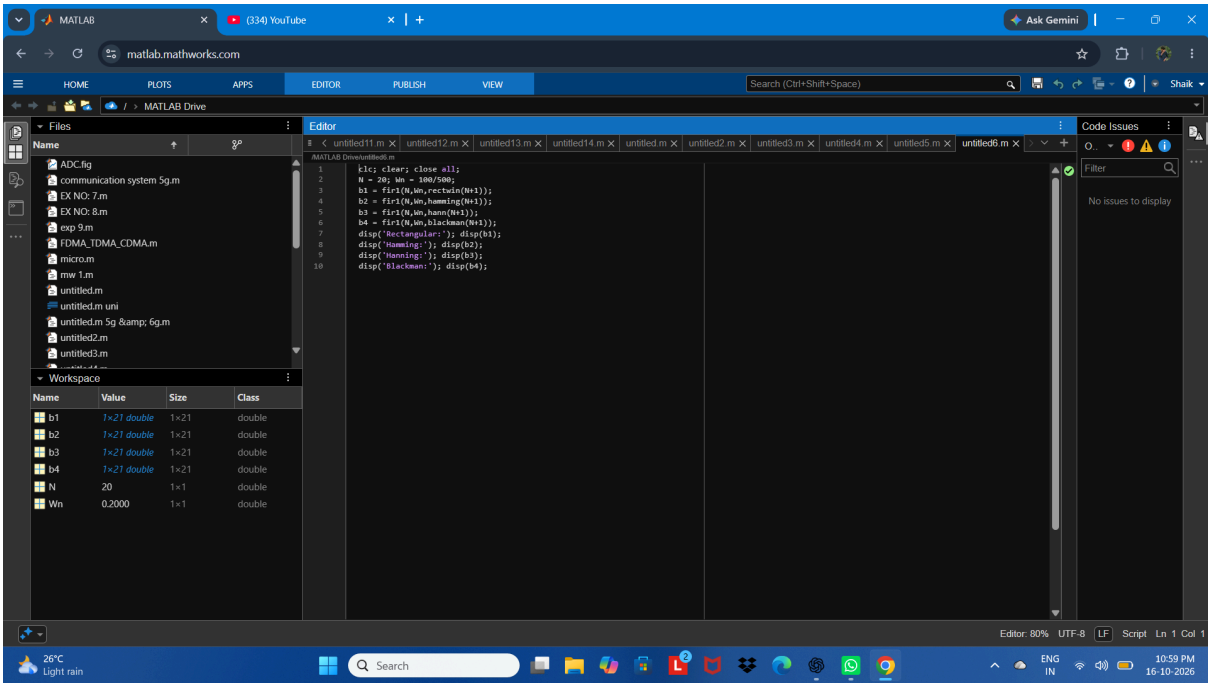
PROGRAM CODES :-

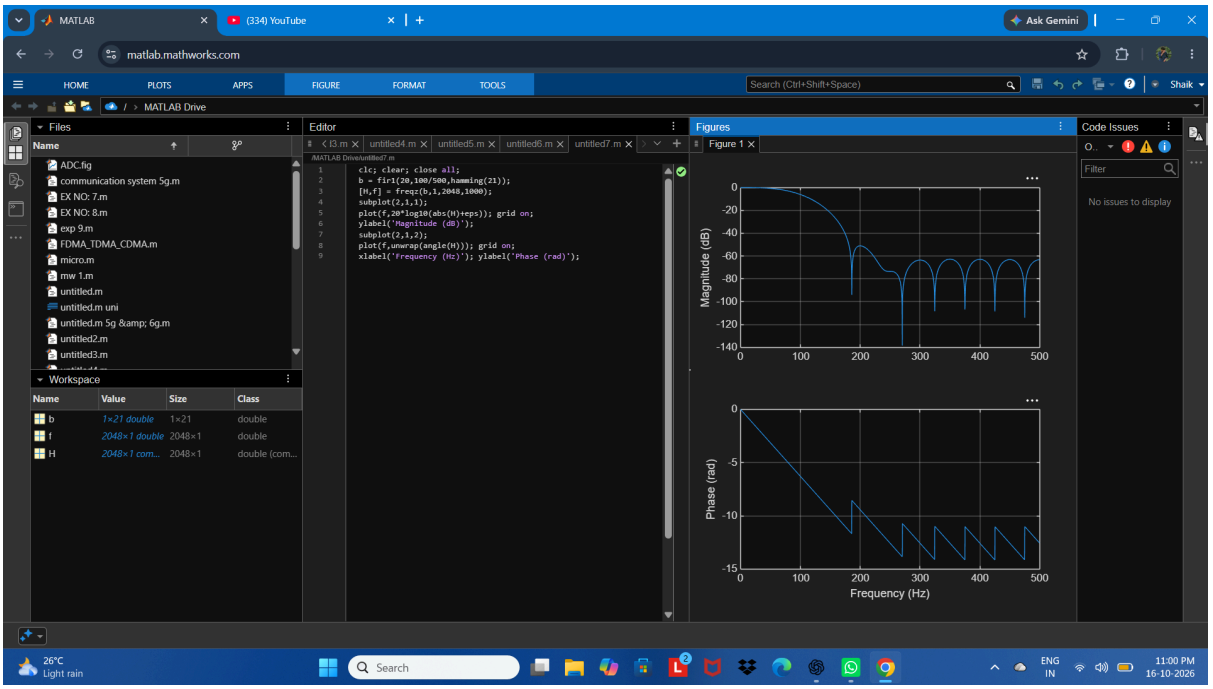
```
clc; clear; close all;
N = 20; Wn = 100/500;
b = fir1(N,Wn,'low',rectwin(N+1));
freqz(b,1,512,1000);
```

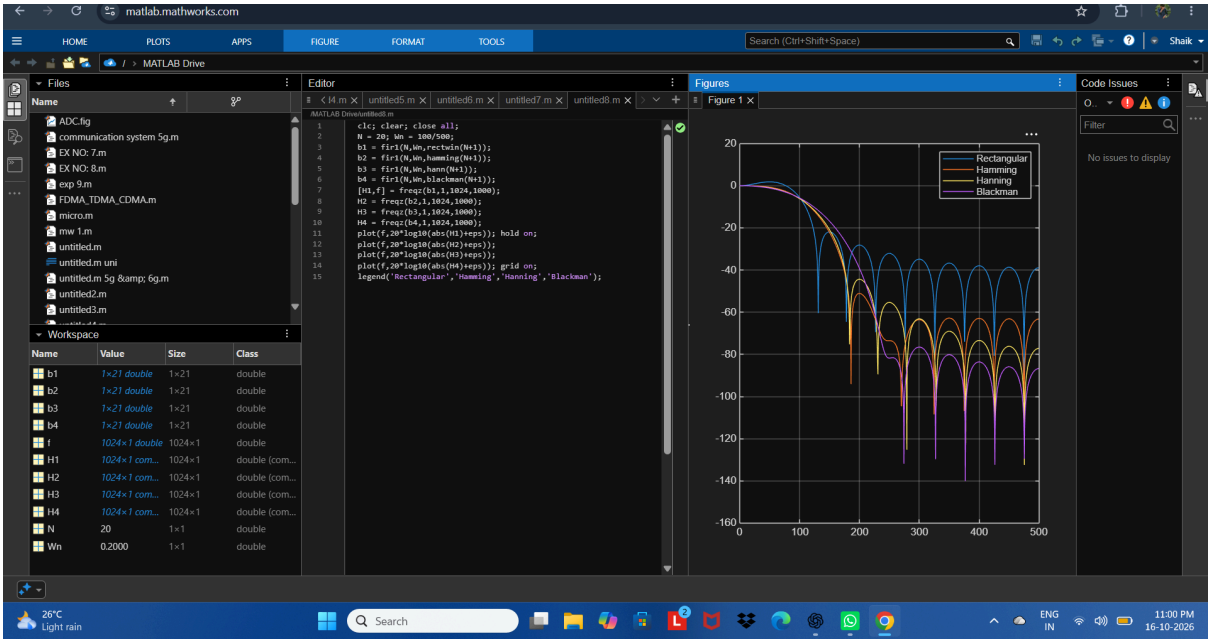
```
clc; clear; close all;
N = 20; Wn = 100/500;
b1 = fir1(N,Wn,rectwin(N+1));
b2 = fir1(N,Wn,hamming(N+1));
b3 = fir1(N,Wn,hann(N+1));
b4 = fir1(N,Wn,blackman(N+1));
disp('Rectangular:'); disp(b1);
disp('Hamming:'); disp(b2);
disp('Hanning:'); disp(b3);
disp('Blackman:'); disp(b4);
```

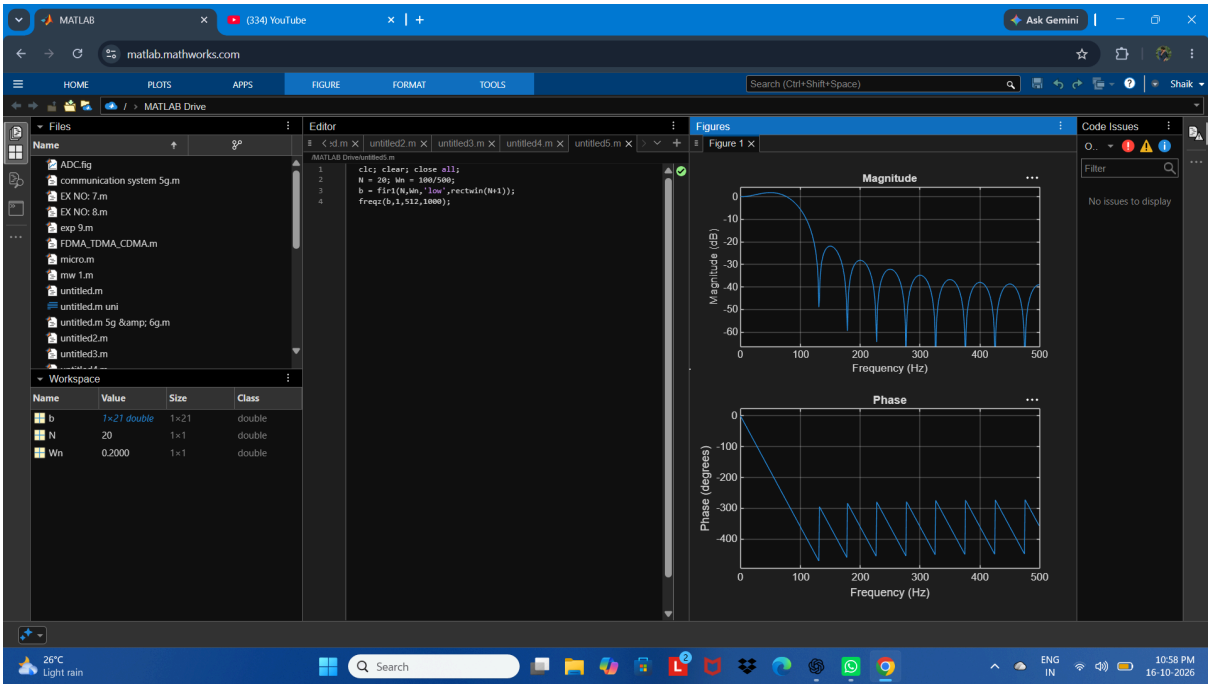
```
clc; clear; close all;
b = fir1(20,100/500,hamming(21));
[H,f] = freqz(b,1,2048,1000);
subplot(2,1,1);
plot(f,20*log10(abs(H)+eps)); grid on;
ylabel('Magnitude (dB)');
subplot(2,1,2);
plot(f,unwrap(angle(H))); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
```

```
clc; clear; close all;
N = 20; Wn = 100/500;
b1 = fir1(N,Wn,rectwin(N+1));
b2 = fir1(N,Wn,hamming(N+1));
b3 = fir1(N,Wn,hann(N+1));
b4 = fir1(N,Wn,blackman(N+1));
[H1,f] = freqz(b1,1,1024,1000);
H2 = freqz(b2,1,1024,1000);
H3 = freqz(b3,1,1024,1000);
H4 = freqz(b4,1,1024,1000);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps));
plot(f,20*log10(abs(H3)+eps));
plot(f,20*log10(abs(H4)+eps)); grid on;
legend('Rectangular','Hamming','Hanning','Blackman');
```


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ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name Value Size Class
b1 1×21 double 1×21 double
b2 1×21 double 1×21 double
b3 1×21 double 1×21 double
b4 1×21 double 1×21 double
N 20 1×1 double
Wn 0.2000 1×1 double
Editor
untitled11.m
untitled12.m
untitled13.m
untitled14.m
untitled.m
untitled2.m
untitled3.m
untitled4.m
untitled5.m
untitled6.m
clc; clear; close all;
N = 20; Wn = 100/500;
b1 = fir1(N,Wn,rectwin(N+1));
b2 = fir1(N,Wn,hamming(N+1));
b3 = fir1(N,Wn,hann(N+1));
b4 = fir1(N,Wn,blackman(N+1));
disp('Rectangular:'); disp(b1);
disp('Hamming:'); disp(b2);
disp('Hanning:'); disp(b3);
disp('Blackman:'); disp(b4);
Code Issues
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EX NO: 8.m
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untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name Value Size Class
b 1×21 double 1×21 double
f 2048×1 double 2048×1 double
H 2048×1 com... 2048×1 double (com...
Editor
t3.m
untitled4.m
untitled5.m
untitled6.m
untitled7.m
clc; clear; close all;
b = fir1(20,100/500,hamming(21));
[H,f] = freqz(b,1,2048,1000);
subplot(2,1,1);
plot(f,20*log10(abs(H)+eps)); grid on;
ylabel('Magnitude (dB)');
subplot(2,1,2);
plot(f,unwrap(angle(H))); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
Figures
Figure 1
Magnitude (dB)
0
-20
-40
-60
-80
-100
-120
-140
0
100
200
300
400
500
Phase (rad)
0
-5
-10
-15
Frequency (Hz)
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